

VetCare AI Model Architectures

Cow Disease Detection Model

- Input Shape: (224, 224, 3)
- Number of Classes: 2 (Healthy, LSD)
- Loss Function: Categorical Crossentropy
- Optimizer: Adam
- Learning Rate: 0.0002
- Metrics: Accuracy

Dog Disease Detection Model

- Input Shape: (224, 224, 3)
- Number of Classes: 3 (Healthy, Fungal, Allergy)
- Loss Function: Categorical Crossentropy
- Optimizer: Adam
- Learning Rate: 0.0002
- Metrics: Accuracy

Model Architectures

Model: "model"

Layer (type)	Output Shape	Param #	Connected to
input_1 (InputLayer)	[(None, 224, 224, 3)]	0	[]
Conv1 (Conv2D)	(None, 112, 112, 32)	864	['input_1[0][0]']
bn_Conv1 (BatchNormalization)	(None, 112, 112, 32)	128	['Conv1[0][0]']
Conv1_relu (ReLU)	(None, 112, 112, 32)	0	['bn_Conv1[0][0]']
expanded_conv_depthwise (DepthwiseConv2D)	(None, 112, 112, 32)	288	['Conv1_relu[0][0]']
expanded_conv_depthwise_BN (BatchNormalization)	(None, 112, 112, 32)	128	['expanded_conv_depthwise[0][0]']
expanded_conv_depthwise_relu (ReLU)	(None, 112, 112, 32)	0	['expanded_conv_depthwise_BN[0][0]']
expanded_conv_project (Conv2D)	(None, 112, 112, 16)	512	['expanded_conv_depthwise_relu[0][0]']
expanded_conv_project_BN (BatchNormalization)	(None, 112, 112, 16)	64	['expanded_conv_project[0][0]']

block_1_expand (Conv2D)	(None, 112, 112, 96)	1536	['expanded_conv_project_BN[0][0]']
block_1_expand_BN (BatchNormalization)	(None, 112, 112, 96)	384	['block_1_expand[0][0]']
block_1_expand_relu (ReLU)	(None, 112, 112, 96)	0	['block_1_expand_BN[0][0]']
block_1_pad (ZeroPadding2D)	(None, 113, 113, 96)	0	['block_1_expand_relu[0][0]']
block_1_depthwise (DepthwiseConv2D)	(None, 56, 56, 96)	864	['block_1_pad[0][0]']
block_1_depthwise_BN (BatchNormalization)	(None, 56, 56, 96)	384	['block_1_depthwise[0][0]']
block_1_depthwise_relu (ReLU)	(None, 56, 56, 96)	0	['block_1_depthwise_BN[0][0]']
block_1_project (Conv2D)	(None, 56, 56, 24)	2304	['block_1_depthwise_relu[0][0]']
block_1_project_BN (BatchNormalization)	(None, 56, 56, 24)	96	['block_1_project[0][0]']
block_2_expand (Conv2D)	(None, 56, 56, 144)	3456	['block_1_project_BN[0][0]']
block_2_expand_BN (BatchNormalization)	(None, 56, 56, 144)	576	['block_2_expand[0][0]']
block_2_expand_relu (ReLU)	(None, 56, 56, 144)	0	['block_2_expand_BN[0][0]']
block_2_depthwise (DepthwiseConv2D)	(None, 56, 56, 144)	1296	['block_2_expand_relu[0][0]']
block_2_depthwise_BN (BatchNormalization)	(None, 56, 56, 144)	576	['block_2_depthwise[0][0]']
block_2_depthwise_relu (ReLU)	(None, 56, 56, 144)	0	['block_2_depthwise_BN[0][0]']
block_2_project (Conv2D)	(None, 56, 56, 24)	3456	['block_2_depthwise_relu[0][0]']
block_2_project_BN (BatchNormalization)	(None, 56, 56, 24)	96	['block_2_project[0][0]']
block_2_add (Add)	(None, 56, 56, 24)	0	['block_1_project_BN[0][0]', 'block_2_project_BN[0][0]']
block_3_expand (Conv2D)	(None, 56, 56, 144)	3456	['block_2_add[0][0]']
block_3_expand_BN (BatchNormalization)	(None, 56, 56, 144)	576	['block_3_expand[0][0]']
block_3_expand_relu (ReLU)	(None, 56, 56, 144)	0	['block_3_expand_BN[0][0]']
block_3_pad (ZeroPadding2D)	(None, 57, 57, 144)	0	['block_3_expand_relu[0][0]']
block_3_depthwise (DepthwiseConv2D)	(None, 28, 28, 144)	1296	['block_3_pad[0][0]']
block_3_depthwise_BN (BatchNormalization)	(None, 28, 28, 144)	576	['block_3_depthwise[0][0]']

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block_3_depthwise_relu (ReLU)	(None, 28, 28, 144)	0	['block_3_depthwise_BN[0][0]']	
block_3_project (Conv2D)	(None, 28, 28, 32)	4608	['block_3_depthwise_relu[0][0]']	
block_3_project_BN (BatchNormalization)	(None, 28, 28, 32)	128	['block_3_project[0][0]']	
block_4_expand (Conv2D)	(None, 28, 28, 192)	6144	['block_3_project_BN[0][0]']	
block_4_expand_BN (BatchNormalization)	(None, 28, 28, 192)	768	['block_4_expand[0][0]']	
block_4_expand_relu (ReLU)	(None, 28, 28, 192)	0	['block_4_expand_BN[0][0]']	
block_4_depthwise (DepthwiseConv2D)	(None, 28, 28, 192)	1728	['block_4_expand_relu[0][0]']	
block_4_depthwise_BN (BatchNormalization)	(None, 28, 28, 192)	768	['block_4_depthwise[0][0]']	
block_4_depthwise_relu (ReLU)	(None, 28, 28, 192)	0	['block_4_depthwise_BN[0][0]']	
block_4_project (Conv2D)	(None, 28, 28, 32)	6144	['block_4_depthwise_relu[0][0]']	
block_4_project_BN (BatchNormalization)	(None, 28, 28, 32)	128	['block_4_project[0][0]']	
block_4_add (Add)	(None, 28, 28, 32)	0	['block_3_project_BN[0][0]', 'block_4_project_BN[0][0]']	
block_5_expand (Conv2D)	(None, 28, 28, 192)	6144	['block_4_add[0][0]']	
block_5_expand_BN (BatchNormalization)	(None, 28, 28, 192)	768	['block_5_expand[0][0]']	
block_5_expand_relu (ReLU)	(None, 28, 28, 192)	0	['block_5_expand_BN[0][0]']	
block_5_depthwise (DepthwiseConv2D)	(None, 28, 28, 192)	1728	['block_5_expand_relu[0][0]']	
block_5_depthwise_BN (BatchNormalization)	(None, 28, 28, 192)	768	['block_5_depthwise[0][0]']	
block_5_depthwise_relu (ReLU)	(None, 28, 28, 192)	0	['block_5_depthwise_BN[0][0]']	
block_5_project (Conv2D)	(None, 28, 28, 32)	6144	['block_5_depthwise_relu[0][0]']	
block_5_project_BN (BatchNormalization)	(None, 28, 28, 32)	128	['block_5_project[0][0]']	
block_5_add (Add)	(None, 28, 28, 32)	0	['block_4_add[0][0]', 'block_5_project_BN[0][0]']	
block_6_expand (Conv2D)	(None, 28, 28, 192)	6144	['block_5_add[0][0]']	
block_6_expand_BN (BatchNormalization)	(None, 28, 28, 192)	768	['block_6_expand[0][0]']	

block_6_expand_relu (ReLU)	(None, 28, 28, 192)	0	['block_6_expand_BN[0][0]']
block_6_pad (ZeroPadding2D)	(None, 29, 29, 192)	0	['block_6_expand_relu[0][0]']
block_6_depthwise (DepthwiseConv2D)	(None, 14, 14, 192)	1728	['block_6_pad[0][0]']
block_6_depthwise_BN (BatchNormalization)	(None, 14, 14, 192)	768	['block_6_depthwise[0][0]']
block_6_depthwise_relu (ReLU)	(None, 14, 14, 192)	0	['block_6_depthwise_BN[0][0]']
block_6_project (Conv2D)	(None, 14, 14, 64)	12288	['block_6_depthwise_relu[0][0]']
block_6_project_BN (BatchNormalization)	(None, 14, 14, 64)	256	['block_6_project[0][0]']
block_7_expand (Conv2D)	(None, 14, 14, 384)	24576	['block_6_project_BN[0][0]']
block_7_expand_BN (BatchNormalization)	(None, 14, 14, 384)	1536	['block_7_expand[0][0]']
block_7_expand_relu (ReLU)	(None, 14, 14, 384)	0	['block_7_expand_BN[0][0]']
block_7_depthwise (DepthwiseConv2D)	(None, 14, 14, 384)	3456	['block_7_expand_relu[0][0]']
block_7_depthwise_BN (BatchNormalization)	(None, 14, 14, 384)	1536	['block_7_depthwise[0][0]']
block_7_depthwise_relu (ReLU)	(None, 14, 14, 384)	0	['block_7_depthwise_BN[0][0]']
block_7_project (Conv2D)	(None, 14, 14, 64)	24576	['block_7_depthwise_relu[0][0]']
block_7_project_BN (BatchNormalization)	(None, 14, 14, 64)	256	['block_7_project[0][0]']
block_7_add (Add)	(None, 14, 14, 64)	0	['block_6_project_BN[0][0]', 'block_7_project_BN[0][0]']
block_8_expand (Conv2D)	(None, 14, 14, 384)	24576	['block_7_add[0][0]']
block_8_expand_BN (BatchNormalization)	(None, 14, 14, 384)	1536	['block_8_expand[0][0]']
block_8_expand_relu (ReLU)	(None, 14, 14, 384)	0	['block_8_expand_BN[0][0]']
block_8_depthwise (DepthwiseConv2D)	(None, 14, 14, 384)	3456	['block_8_expand_relu[0][0]']
block_8_depthwise_BN (BatchNormalization)	(None, 14, 14, 384)	1536	['block_8_depthwise[0][0]']
block_8_depthwise_relu (ReLU)	(None, 14, 14, 384)	0	['block_8_depthwise_BN[0][0]']
block_8_project (Conv2D)	(None, 14, 14, 64)	24576	['block_8_depthwise_relu[0][0]']
block_8_project_BN (BatchNormalization)	(None, 14, 14, 64)	256	['block_8_project[0][0]']

block_8_add (Add)	(None, 14, 14, 64)	0	['block_7_add[0][0]', 'block_8_project_BN[0][0]']
block_9_expand (Conv2D)	(None, 14, 14, 384)	24576	['block_8_add[0][0]']
block_9_expand_BN (Batch Normalization)	(None, 14, 14, 384)	1536	['block_9_expand[0][0]']
block_9_expand_relu (ReLU)	(None, 14, 14, 384)	0	['block_9_expand_BN[0][0]']
block_9_depthwise (Depthwise Conv2D)	(None, 14, 14, 384)	3456	['block_9_expand_relu[0][0]']
block_9_depthwise_BN (Batch Normalization)	(None, 14, 14, 384)	1536	['block_9_depthwise[0][0]']
block_9_depthwise_relu (ReLU)	(None, 14, 14, 384)	0	['block_9_depthwise_BN[0][0]']
block_9_project (Conv2D)	(None, 14, 14, 64)	24576	['block_9_depthwise_relu[0][0]']
block_9_project_BN (Batch Normalization)	(None, 14, 14, 64)	256	['block_9_project[0][0]']
block_9_add (Add)	(None, 14, 14, 64)	0	['block_8_add[0][0]', 'block_9_project_BN[0][0]']
block_10_expand (Conv2D)	(None, 14, 14, 384)	24576	['block_9_add[0][0]']
block_10_expand_BN (Batch Normalization)	(None, 14, 14, 384)	1536	['block_10_expand[0][0]']
block_10_expand_relu (ReLU)	(None, 14, 14, 384)	0	['block_10_expand_BN[0][0]']
block_10_depthwise (Depthwise Conv2D)	(None, 14, 14, 384)	3456	['block_10_expand_relu[0][0]']
block_10_depthwise_BN (Batch Normalization)	(None, 14, 14, 384)	1536	['block_10_depthwise[0][0]']
block_10_depthwise_relu (ReLU)	(None, 14, 14, 384)	0	['block_10_depthwise_BN[0][0]']
block_10_project (Conv2D)	(None, 14, 14, 96)	36864	['block_10_depthwise_relu[0][0]']
block_10_project_BN (Batch Normalization)	(None, 14, 14, 96)	384	['block_10_project[0][0]']
block_11_expand (Conv2D)	(None, 14, 14, 576)	55296	['block_10_project_BN[0][0]']
block_11_expand_BN (Batch Normalization)	(None, 14, 14, 576)	2304	['block_11_expand[0][0]']
block_11_expand_relu (ReLU)	(None, 14, 14, 576)	0	['block_11_expand_BN[0][0]']
block_11_depthwise (Depthwise Conv2D)	(None, 14, 14, 576)	5184	['block_11_expand_relu[0][0]']
block_11_depthwise_BN (Batch Normalization)	(None, 14, 14, 576)	2304	['block_11_depthwise[0][0]']

block_11_depthwise_relu (ReLU)	(None, 14, 14, 576)	0	['block_11_depthwise_BN[0][0]']
block_11_project (Conv2D)	(None, 14, 14, 96)	55296	['block_11_depthwise_relu[0][0]']
block_11_project_BN (Batch Normalization)	(None, 14, 14, 96)	384	['block_11_project[0][0]']
block_11_add (Add)	(None, 14, 14, 96)	0	['block_10_project_BN[0][0]', 'block_11_project_BN[0][0]']
block_12_expand (Conv2D)	(None, 14, 14, 576)	55296	['block_11_add[0][0]']
block_12_expand_BN (Batch Normalization)	(None, 14, 14, 576)	2304	['block_12_expand[0][0]']
block_12_expand_relu (ReLU)	(None, 14, 14, 576)	0	['block_12_expand_BN[0][0]']
block_12_depthwise (DepthwiseConv2D)	(None, 14, 14, 576)	5184	['block_12_expand_relu[0][0]']
block_12_depthwise_BN (Batch Normalization)	(None, 14, 14, 576)	2304	['block_12_depthwise[0][0]']
block_12_depthwise_relu (ReLU)	(None, 14, 14, 576)	0	['block_12_depthwise_BN[0][0]']
block_12_project (Conv2D)	(None, 14, 14, 96)	55296	['block_12_depthwise_relu[0][0]']
block_12_project_BN (Batch Normalization)	(None, 14, 14, 96)	384	['block_12_project[0][0]']
block_12_add (Add)	(None, 14, 14, 96)	0	['block_11_add[0][0]', 'block_12_project_BN[0][0]']
block_13_expand (Conv2D)	(None, 14, 14, 576)	55296	['block_12_add[0][0]']
block_13_expand_BN (Batch Normalization)	(None, 14, 14, 576)	2304	['block_13_expand[0][0]']
block_13_expand_relu (ReLU)	(None, 14, 14, 576)	0	['block_13_expand_BN[0][0]']
block_13_pad (ZeroPadding2D)	(None, 15, 15, 576)	0	['block_13_expand_relu[0][0]']
block_13_depthwise (DepthwiseConv2D)	(None, 7, 7, 576)	5184	['block_13_pad[0][0]']
block_13_depthwise_BN (Batch Normalization)	(None, 7, 7, 576)	2304	['block_13_depthwise[0][0]']
block_13_depthwise_relu (ReLU)	(None, 7, 7, 576)	0	['block_13_depthwise_BN[0][0]']
block_13_project (Conv2D)	(None, 7, 7, 160)	92160	['block_13_depthwise_relu[0][0]']
block_13_project_BN (Batch Normalization)	(None, 7, 7, 160)	640	['block_13_project[0][0]']
block_14_expand (Conv2D)	(None, 7, 7, 960)	153600	['block_13_project_BN[0][0]']

block_14_expand_BN (Batch Normalization)	(None, 7, 7, 960)	3840	['block_14_expand[0][0]']
block_14_expand_relu (ReLU)	(None, 7, 7, 960)	0	['block_14_expand_BN[0][0]']
block_14_depthwise (DepthwiseConv2D)	(None, 7, 7, 960)	8640	['block_14_expand_relu[0][0]']
block_14_depthwise_BN (Batch Normalization)	(None, 7, 7, 960)	3840	['block_14_depthwise[0][0]']
block_14_depthwise_relu (ReLU)	(None, 7, 7, 960)	0	['block_14_depthwise_BN[0][0]']
block_14_project (Conv2D)	(None, 7, 7, 160)	153600	['block_14_depthwise_relu[0][0]']
block_14_project_BN (Batch Normalization)	(None, 7, 7, 160)	640	['block_14_project[0][0]']
block_14_add (Add)	(None, 7, 7, 160)	0	['block_13_project_BN[0][0]', 'block_14_project_BN[0][0]']
block_15_expand (Conv2D)	(None, 7, 7, 960)	153600	['block_14_add[0][0]']
block_15_expand_BN (Batch Normalization)	(None, 7, 7, 960)	3840	['block_15_expand[0][0]']
block_15_expand_relu (ReLU)	(None, 7, 7, 960)	0	['block_15_expand_BN[0][0]']
block_15_depthwise (DepthwiseConv2D)	(None, 7, 7, 960)	8640	['block_15_expand_relu[0][0]']
block_15_depthwise_BN (Batch Normalization)	(None, 7, 7, 960)	3840	['block_15_depthwise[0][0]']
block_15_depthwise_relu (ReLU)	(None, 7, 7, 960)	0	['block_15_depthwise_BN[0][0]']
block_15_project (Conv2D)	(None, 7, 7, 160)	153600	['block_15_depthwise_relu[0][0]']
block_15_project_BN (Batch Normalization)	(None, 7, 7, 160)	640	['block_15_project[0][0]']
block_15_add (Add)	(None, 7, 7, 160)	0	['block_14_add[0][0]', 'block_15_project_BN[0][0]']
block_16_expand (Conv2D)	(None, 7, 7, 960)	153600	['block_15_add[0][0]']
block_16_expand_BN (Batch Normalization)	(None, 7, 7, 960)	3840	['block_16_expand[0][0]']
block_16_expand_relu (ReLU)	(None, 7, 7, 960)	0	['block_16_expand_BN[0][0]']
block_16_depthwise (DepthwiseConv2D)	(None, 7, 7, 960)	8640	['block_16_expand_relu[0][0]']
block_16_depthwise_BN (Batch Normalization)	(None, 7, 7, 960)	3840	['block_16_depthwise[0][0]']
block_16_depthwise_relu (ReLU)	(None, 7, 7, 960)	0	['block_16_depthwise_BN[0][0]']

block_16_project (Conv2D)	(None, 7, 7, 320)	307200	['block_16_depthwise_relu[0][0]']
block_16_project_BN (Batch Normalization)	(None, 7, 7, 320)	1280	['block_16_project[0][0]']
Conv_1 (Conv2D)	(None, 7, 7, 1280)	409600	['block_16_project_BN[0][0]']
Conv_1_bn (BatchNormalization)	(None, 7, 7, 1280)	5120	['Conv_1[0][0]']
out_relu (ReLU)	(None, 7, 7, 1280)	0	['Conv_1_bn[0][0]']
global_average_pooling2d (GlobalAveragePooling2D)	(None, 1280)	0	['out_relu[0][0]']
dense (Dense)	(None, 256)	327936	['global_average_pooling2d[0][0]']
dropout (Dropout)	(None, 256)	0	['dense[0][0]']
dense_1 (Dense)	(None, 128)	32896	['dropout[0][0]']
dense_2 (Dense)	(None, 2)	258	['dense_1[0][0]']

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 Total params: 2619074 (9.99 MB)
 Trainable params: 361090 (1.38 MB)
 Non-trainable params: 2257984 (8.61 MB)

Model: "model_1"

Layer (type)	Output Shape	Param #	Connected to
input_2 (InputLayer)	[(None, 224, 224, 3)]	0	[]
Conv1 (Conv2D)	(None, 112, 112, 32)	864	['input_2[0][0]']
bn_Conv1 (BatchNormalization)	(None, 112, 112, 32)	128	['Conv1[0][0]']
Conv1_relu (ReLU)	(None, 112, 112, 32)	0	['bn_Conv1[0][0]']
expanded_conv_depthwise (DepthwiseConv2D)	(None, 112, 112, 32)	288	['Conv1_relu[0][0]']
expanded_conv_depthwise_BN (BatchNormalization)	(None, 112, 112, 32)	128	['expanded_conv_depthwise[0][0]']
expanded_conv_depthwise_relu (ReLU)	(None, 112, 112, 32)	0	['expanded_conv_depthwise_BN[0][0]']
expanded_conv_project (Conv2D)	(None, 112, 112, 16)	512	['expanded_conv_depthwise_relu[0][0]']
expanded_conv_project_BN (BatchNormalization)	(None, 112, 112, 16)	64	['expanded_conv_project[0][0]']
block_1_expand (Conv2D)	(None, 112, 112, 96)	1536	['expanded_conv_project_BN[0][0]']
block_1_expand_BN (BatchNormalization)	(None, 112, 112, 96)	384	['block_1_expand[0][0]']

block_1_expand_relu (ReLU)	(None, 112, 112, 96)	0	['block_1_expand_BN[0][0]']
block_1_pad (ZeroPadding2D)	(None, 113, 113, 96)	0	['block_1_expand_relu[0][0]']
block_1_depthwise (DepthwiseConv2D)	(None, 56, 56, 96)	864	['block_1_pad[0][0]']
block_1_depthwise_BN (BatchNormalization)	(None, 56, 56, 96)	384	['block_1_depthwise[0][0]']
block_1_depthwise_relu (ReLU)	(None, 56, 56, 96)	0	['block_1_depthwise_BN[0][0]']
block_1_project (Conv2D)	(None, 56, 56, 24)	2304	['block_1_depthwise_relu[0][0]']
block_1_project_BN (BatchNormalization)	(None, 56, 56, 24)	96	['block_1_project[0][0]']
block_2_expand (Conv2D)	(None, 56, 56, 144)	3456	['block_1_project_BN[0][0]']
block_2_expand_BN (BatchNormalization)	(None, 56, 56, 144)	576	['block_2_expand[0][0]']
block_2_expand_relu (ReLU)	(None, 56, 56, 144)	0	['block_2_expand_BN[0][0]']
block_2_depthwise (DepthwiseConv2D)	(None, 56, 56, 144)	1296	['block_2_expand_relu[0][0]']
block_2_depthwise_BN (BatchNormalization)	(None, 56, 56, 144)	576	['block_2_depthwise[0][0]']
block_2_depthwise_relu (ReLU)	(None, 56, 56, 144)	0	['block_2_depthwise_BN[0][0]']
block_2_project (Conv2D)	(None, 56, 56, 24)	3456	['block_2_depthwise_relu[0][0]']
block_2_project_BN (BatchNormalization)	(None, 56, 56, 24)	96	['block_2_project[0][0]']
block_2_add (Add)	(None, 56, 56, 24)	0	['block_1_project_BN[0][0]', 'block_2_project_BN[0][0]']
block_3_expand (Conv2D)	(None, 56, 56, 144)	3456	['block_2_add[0][0]']
block_3_expand_BN (BatchNormalization)	(None, 56, 56, 144)	576	['block_3_expand[0][0]']
block_3_expand_relu (ReLU)	(None, 56, 56, 144)	0	['block_3_expand_BN[0][0]']
block_3_pad (ZeroPadding2D)	(None, 57, 57, 144)	0	['block_3_expand_relu[0][0]']
block_3_depthwise (DepthwiseConv2D)	(None, 28, 28, 144)	1296	['block_3_pad[0][0]']
block_3_depthwise_BN (BatchNormalization)	(None, 28, 28, 144)	576	['block_3_depthwise[0][0]']
block_3_depthwise_relu (ReLU)	(None, 28, 28, 144)	0	['block_3_depthwise_BN[0][0]']
block_3_project (Conv2D)	(None, 28, 28, 32)	4608	['block_3_depthwise_relu[0][0]']

block_3_project_BN (BatchNormalization)	(None, 28, 28, 32)	128	['block_3_project[0][0]']
block_4_expand (Conv2D)	(None, 28, 28, 192)	6144	['block_3_project_BN[0][0]']
block_4_expand_BN (BatchNormalization)	(None, 28, 28, 192)	768	['block_4_expand[0][0]']
block_4_expand_relu (ReLU)	(None, 28, 28, 192)	0	['block_4_expand_BN[0][0]']
block_4_depthwise (DepthwiseConv2D)	(None, 28, 28, 192)	1728	['block_4_expand_relu[0][0]']
block_4_depthwise_BN (BatchNormalization)	(None, 28, 28, 192)	768	['block_4_depthwise[0][0]']
block_4_depthwise_relu (ReLU)	(None, 28, 28, 192)	0	['block_4_depthwise_BN[0][0]']
block_4_project (Conv2D)	(None, 28, 28, 32)	6144	['block_4_depthwise_relu[0][0]']
block_4_project_BN (BatchNormalization)	(None, 28, 28, 32)	128	['block_4_project[0][0]']
block_4_add (Add)	(None, 28, 28, 32)	0	['block_3_project_BN[0][0]', 'block_4_project_BN[0][0]']
block_5_expand (Conv2D)	(None, 28, 28, 192)	6144	['block_4_add[0][0]']
block_5_expand_BN (BatchNormalization)	(None, 28, 28, 192)	768	['block_5_expand[0][0]']
block_5_expand_relu (ReLU)	(None, 28, 28, 192)	0	['block_5_expand_BN[0][0]']
block_5_depthwise (DepthwiseConv2D)	(None, 28, 28, 192)	1728	['block_5_expand_relu[0][0]']
block_5_depthwise_BN (BatchNormalization)	(None, 28, 28, 192)	768	['block_5_depthwise[0][0]']
block_5_depthwise_relu (ReLU)	(None, 28, 28, 192)	0	['block_5_depthwise_BN[0][0]']
block_5_project (Conv2D)	(None, 28, 28, 32)	6144	['block_5_depthwise_relu[0][0]']
block_5_project_BN (BatchNormalization)	(None, 28, 28, 32)	128	['block_5_project[0][0]']
block_5_add (Add)	(None, 28, 28, 32)	0	['block_4_add[0][0]', 'block_5_project_BN[0][0]']
block_6_expand (Conv2D)	(None, 28, 28, 192)	6144	['block_5_add[0][0]']
block_6_expand_BN (BatchNormalization)	(None, 28, 28, 192)	768	['block_6_expand[0][0]']
block_6_expand_relu (ReLU)	(None, 28, 28, 192)	0	['block_6_expand_BN[0][0]']
block_6_pad (ZeroPadding2D)	(None, 29, 29, 192)	0	['block_6_expand_relu[0][0]']
block_6_depthwise (DepthwiseConv2D)	(None, 14, 14, 192)	1728	['block_6_pad[0][0]']

block_6_depthwise_BN (Batch Normalization)	(None, 14, 14, 192)	768	['block_6_depthwise[0][0]']
block_6_depthwise_relu (ReLU)	(None, 14, 14, 192)	0	['block_6_depthwise_BN[0][0]']
block_6_project (Conv2D)	(None, 14, 14, 64)	12288	['block_6_depthwise_relu[0][0]']
block_6_project_BN (Batch Normalization)	(None, 14, 14, 64)	256	['block_6_project[0][0]']
block_7_expand (Conv2D)	(None, 14, 14, 384)	24576	['block_6_project_BN[0][0]']
block_7_expand_BN (Batch Normalization)	(None, 14, 14, 384)	1536	['block_7_expand[0][0]']
block_7_expand_relu (ReLU)	(None, 14, 14, 384)	0	['block_7_expand_BN[0][0]']
block_7_depthwise (Depthwise Conv2D)	(None, 14, 14, 384)	3456	['block_7_expand_relu[0][0]']
block_7_depthwise_BN (Batch Normalization)	(None, 14, 14, 384)	1536	['block_7_depthwise[0][0]']
block_7_depthwise_relu (ReLU)	(None, 14, 14, 384)	0	['block_7_depthwise_BN[0][0]']
block_7_project (Conv2D)	(None, 14, 14, 64)	24576	['block_7_depthwise_relu[0][0]']
block_7_project_BN (Batch Normalization)	(None, 14, 14, 64)	256	['block_7_project[0][0]']
block_7_add (Add)	(None, 14, 14, 64)	0	['block_6_project_BN[0][0]', 'block_7_project_BN[0][0]']
block_8_expand (Conv2D)	(None, 14, 14, 384)	24576	['block_7_add[0][0]']
block_8_expand_BN (Batch Normalization)	(None, 14, 14, 384)	1536	['block_8_expand[0][0]']
block_8_expand_relu (ReLU)	(None, 14, 14, 384)	0	['block_8_expand_BN[0][0]']
block_8_depthwise (Depthwise Conv2D)	(None, 14, 14, 384)	3456	['block_8_expand_relu[0][0]']
block_8_depthwise_BN (Batch Normalization)	(None, 14, 14, 384)	1536	['block_8_depthwise[0][0]']
block_8_depthwise_relu (ReLU)	(None, 14, 14, 384)	0	['block_8_depthwise_BN[0][0]']
block_8_project (Conv2D)	(None, 14, 14, 64)	24576	['block_8_depthwise_relu[0][0]']
block_8_project_BN (Batch Normalization)	(None, 14, 14, 64)	256	['block_8_project[0][0]']
block_8_add (Add)	(None, 14, 14, 64)	0	['block_7_add[0][0]', 'block_8_project_BN[0][0]']
block_9_expand (Conv2D)	(None, 14, 14, 384)	24576	['block_8_add[0][0]']
block_9_expand_BN (Batch Normalization)	(None, 14, 14, 384)	1536	['block_9_expand[0][0]']

rmalization)				
block_9_expand_relu (ReLU)	(None, 14, 14, 384)	0	['block_9_expand_BN[0][0]']	
block_9_depthwise (DepthwiseConv2D)	(None, 14, 14, 384)	3456	['block_9_expand_relu[0][0]']	
block_9_depthwise_BN (BatchNormalization)	(None, 14, 14, 384)	1536	['block_9_depthwise[0][0]']	
block_9_depthwise_relu (ReLU)	(None, 14, 14, 384)	0	['block_9_depthwise_BN[0][0]']	
block_9_project (Conv2D)	(None, 14, 14, 64)	24576	['block_9_depthwise_relu[0][0]']	
block_9_project_BN (BatchNormalization)	(None, 14, 14, 64)	256	['block_9_project[0][0]']	
block_9_add (Add)	(None, 14, 14, 64)	0	['block_8_add[0][0]', 'block_9_project_BN[0][0]']	
block_10_expand (Conv2D)	(None, 14, 14, 384)	24576	['block_9_add[0][0]']	
block_10_expand_BN (BatchNormalization)	(None, 14, 14, 384)	1536	['block_10_expand[0][0]']	
block_10_expand_relu (ReLU)	(None, 14, 14, 384)	0	['block_10_expand_BN[0][0]']	
block_10_depthwise (DepthwiseConv2D)	(None, 14, 14, 384)	3456	['block_10_expand_relu[0][0]']	
block_10_depthwise_BN (BatchNormalization)	(None, 14, 14, 384)	1536	['block_10_depthwise[0][0]']	
block_10_depthwise_relu (ReLU)	(None, 14, 14, 384)	0	['block_10_depthwise_BN[0][0]']	
block_10_project (Conv2D)	(None, 14, 14, 96)	36864	['block_10_depthwise_relu[0][0]']	
block_10_project_BN (BatchNormalization)	(None, 14, 14, 96)	384	['block_10_project[0][0]']	
block_11_expand (Conv2D)	(None, 14, 14, 576)	55296	['block_10_project_BN[0][0]']	
block_11_expand_BN (BatchNormalization)	(None, 14, 14, 576)	2304	['block_11_expand[0][0]']	
block_11_expand_relu (ReLU)	(None, 14, 14, 576)	0	['block_11_expand_BN[0][0]']	
block_11_depthwise (DepthwiseConv2D)	(None, 14, 14, 576)	5184	['block_11_expand_relu[0][0]']	
block_11_depthwise_BN (BatchNormalization)	(None, 14, 14, 576)	2304	['block_11_depthwise[0][0]']	
block_11_depthwise_relu (ReLU)	(None, 14, 14, 576)	0	['block_11_depthwise_BN[0][0]']	
block_11_project (Conv2D)	(None, 14, 14, 96)	55296	['block_11_depthwise_relu[0][0]']	
block_11_project_BN (Batch	(None, 14, 14, 96)	384	['block_11_project[0][0]']	

Normalization)

block_11_add (Add)	(None, 14, 14, 96)	0	['block_10_project_BN[0][0]', 'block_11_project_BN[0][0]']
block_12_expand (Conv2D)	(None, 14, 14, 576)	55296	['block_11_add[0][0]']
block_12_expand_BN (BatchNormalization)	(None, 14, 14, 576)	2304	['block_12_expand[0][0]']
block_12_expand_relu (ReLU)	(None, 14, 14, 576)	0	['block_12_expand_BN[0][0]']
block_12_depthwise (DepthwiseConv2D)	(None, 14, 14, 576)	5184	['block_12_expand_relu[0][0]']
block_12_depthwise_BN (BatchNormalization)	(None, 14, 14, 576)	2304	['block_12_depthwise[0][0]']
block_12_depthwise_relu (ReLU)	(None, 14, 14, 576)	0	['block_12_depthwise_BN[0][0]']
block_12_project (Conv2D)	(None, 14, 14, 96)	55296	['block_12_depthwise_relu[0][0]']
block_12_project_BN (BatchNormalization)	(None, 14, 14, 96)	384	['block_12_project[0][0]']
block_12_add (Add)	(None, 14, 14, 96)	0	['block_11_add[0][0]', 'block_12_project_BN[0][0]']
block_13_expand (Conv2D)	(None, 14, 14, 576)	55296	['block_12_add[0][0]']
block_13_expand_BN (BatchNormalization)	(None, 14, 14, 576)	2304	['block_13_expand[0][0]']
block_13_expand_relu (ReLU)	(None, 14, 14, 576)	0	['block_13_expand_BN[0][0]']
block_13_pad (ZeroPadding2D)	(None, 15, 15, 576)	0	['block_13_expand_relu[0][0]']
block_13_depthwise (DepthwiseConv2D)	(None, 7, 7, 576)	5184	['block_13_pad[0][0]']
block_13_depthwise_BN (BatchNormalization)	(None, 7, 7, 576)	2304	['block_13_depthwise[0][0]']
block_13_depthwise_relu (ReLU)	(None, 7, 7, 576)	0	['block_13_depthwise_BN[0][0]']
block_13_project (Conv2D)	(None, 7, 7, 160)	92160	['block_13_depthwise_relu[0][0]']
block_13_project_BN (BatchNormalization)	(None, 7, 7, 160)	640	['block_13_project[0][0]']
block_14_expand (Conv2D)	(None, 7, 7, 960)	153600	['block_13_project_BN[0][0]']
block_14_expand_BN (BatchNormalization)	(None, 7, 7, 960)	3840	['block_14_expand[0][0]']
block_14_expand_relu (ReLU)	(None, 7, 7, 960)	0	['block_14_expand_BN[0][0]']
block_14_depthwise (DepthwiseConv2D)	(None, 7, 7, 960)	8640	['block_14_expand_relu[0][0]']

iseConv2D)				
block_14_depthwise_BN (Batch Normalization)	(None, 7, 7, 960)	3840	['block_14_depthwise[0][0]']	
block_14_depthwise_relu (ReLU)	(None, 7, 7, 960)	0	['block_14_depthwise_BN[0][0]']	
block_14_project (Conv2D)	(None, 7, 7, 160)	153600	['block_14_depthwise_relu[0][0]']	
block_14_project_BN (Batch Normalization)	(None, 7, 7, 160)	640	['block_14_project[0][0]']	
block_14_add (Add)	(None, 7, 7, 160)	0	['block_13_project_BN[0][0]', 'block_14_project_BN[0][0]']	
block_15_expand (Conv2D)	(None, 7, 7, 960)	153600	['block_14_add[0][0]']	
block_15_expand_BN (Batch Normalization)	(None, 7, 7, 960)	3840	['block_15_expand[0][0]']	
block_15_expand_relu (ReLU)	(None, 7, 7, 960)	0	['block_15_expand_BN[0][0]']	
block_15_depthwise (Depthwise Conv2D)	(None, 7, 7, 960)	8640	['block_15_expand_relu[0][0]']	
block_15_depthwise_BN (Batch Normalization)	(None, 7, 7, 960)	3840	['block_15_depthwise[0][0]']	
block_15_depthwise_relu (ReLU)	(None, 7, 7, 960)	0	['block_15_depthwise_BN[0][0]']	
block_15_project (Conv2D)	(None, 7, 7, 160)	153600	['block_15_depthwise_relu[0][0]']	
block_15_project_BN (Batch Normalization)	(None, 7, 7, 160)	640	['block_15_project[0][0]']	
block_15_add (Add)	(None, 7, 7, 160)	0	['block_14_add[0][0]', 'block_15_project_BN[0][0]']	
block_16_expand (Conv2D)	(None, 7, 7, 960)	153600	['block_15_add[0][0]']	
block_16_expand_BN (Batch Normalization)	(None, 7, 7, 960)	3840	['block_16_expand[0][0]']	
block_16_expand_relu (ReLU)	(None, 7, 7, 960)	0	['block_16_expand_BN[0][0]']	
block_16_depthwise (Depthwise Conv2D)	(None, 7, 7, 960)	8640	['block_16_expand_relu[0][0]']	
block_16_depthwise_BN (Batch Normalization)	(None, 7, 7, 960)	3840	['block_16_depthwise[0][0]']	
block_16_depthwise_relu (ReLU)	(None, 7, 7, 960)	0	['block_16_depthwise_BN[0][0]']	
block_16_project (Conv2D)	(None, 7, 7, 320)	307200	['block_16_depthwise_relu[0][0]']	
block_16_project_BN (Batch Normalization)	(None, 7, 7, 320)	1280	['block_16_project[0][0]']	

Conv_1 (Conv2D)	(None, 7, 7, 1280)	409600	['block_16_project_BN[0][0]']
Conv_1_bn (BatchNormalization)	(None, 7, 7, 1280)	5120	['Conv_1[0][0]']
out_relu (ReLU)	(None, 7, 7, 1280)	0	['Conv_1_bn[0][0]']
global_average_pooling2d_1 (GlobalAveragePooling2D)	(None, 1280)	0	['out_relu[0][0]']
dense_3 (Dense)	(None, 256)	327936	['global_average_pooling2d_1[0][0]']
dropout_1 (Dropout)	(None, 256)	0	['dense_3[0][0]']
dense_4 (Dense)	(None, 128)	32896	['dropout_1[0][0]']
dense_5 (Dense)	(None, 3)	387	['dense_4[0][0]']

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Total params: 2619203 (9.99 MB)

Trainable params: 361219 (1.38 MB)

Non-trainable params: 2257984 (8.61 MB)

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